

Cognitva Healthcare AI Challenge 2026

Problem 1: AI-Based Healthcare Data De-Identification Engine

Objective

Build an AI-powered system capable of detecting and anonymizing Protected Health Information (PHI) from clinical documents while preserving medical context and semantic integrity.

Core Requirements

The solution must:

- Accept PDF, text, and scanned documents (via OCR).
- Detect PHI categories including name, age, phone number, email, address, hospital ID, dates, doctor name, and organization name.
- Preserve medical values, diagnoses, lab parameters, and clinical measurements.
- Output a redacted document and a structured JSON containing detected PHI.
- Provide confidence scores and anonymization quality metrics.

Advanced/Bonus Requirements

- Support multilingual anonymization (English + 1 Indian language).
- Provide explainability layer for detected PHI.
- Handle OCR noise robustness.

Dataset Guidance

Participants may use the following publicly available datasets:

- MIMIC-IV dataset (PhysioNet).
- i2b2 De-identification Challenge datasets (academic license required).
- Synthetic datasets generated using Faker or LLM-based tools.
- Public prescription image samples for OCR testing.